

Precept 6

ORF 309

Goals

- Continuous distributions (density, expectation, variance)
- The density (pdf) is the derivative of the cumulative distribution function (cdf)
- How to use the law of total probability and the law of total expectation (same as tower property)
- Joint distribution, marginal distribution, conditional distribution for continuous random variables
- Meaning of $\mathbb{P}[X \in dx|Y = y]$ and comparison with the discrete case.

Reminder (Continuous random variables, joint and conditional densities)

Let $X : \Omega \mapsto \mathbb{R}$ be a continuous random variable. Its distribution is characterized by the CDF (Cumulative Distribution Function) $F : \mathbb{R} \mapsto [0, 1]$:

$$\forall x \in \mathbb{R}, F(x) = \mathbb{P}(X \leq x)$$

The density function f is defined as the derivative of the CDF :

$$f(x) dx = dF(x) = \mathbb{P}(X \in dx), \quad \text{or} \quad f(x) = \frac{d}{dx} F(x) = F'(x).$$

The expectation is defined as:

$$\mathbb{E}[X] = \int x \mathbb{P}(X \in dx) = \int x f(x) dx$$

and the variance can be calculated as $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ thanks to the formula :

$$\mathbb{E}[X^2] = \int x^2 \mathbb{P}(X \in dx) = \int x^2 f(x) dx$$

(Law Of The Unconscious Statistician - Continuous version) For a function φ , we have :

$$\mathbb{E}[\varphi(X)] = \int \varphi(x) \mathbb{P}(X \in dx) = \int \varphi(x) f(x) dx \quad (\text{LOTUS})$$

Let X and Y be two random variables.

$$\begin{aligned}
- f_{X,Y}(x,y) &= \frac{\partial^2}{\partial x \partial y} \mathbb{P}(X \leq x, Y \leq y) && \text{(joint density of } X \text{ and } Y) \\
- \mathbb{E}[\varphi(X,Y)] &= \int \varphi(x,y) \mathbb{P}(X \in dx, Y \in dy) = \int \varphi(x,y) f(x,y) dx dy && \text{(expectation of } \varphi(X,Y)) \\
- f_X(x) &= \int f_{X,Y}(x,y) dy && \text{(marginal density of } X) \\
- \mathbb{P}(X \in dx | Y = y) &= \frac{\mathbb{P}(X \in dx, Y \in dy)}{\mathbb{P}(Y \in dy)} && \text{(conditional distribution of } X \text{ given } Y = y) \\
- \mathbb{P}(X \in dx | Y = y) &= \frac{f(x,y)}{\int f(\tilde{x},y) d\tilde{x}} dx = f_{X|Y}(x|y) dx && \text{(conditional density of } X \text{ given } Y = y) \\
- \mathbb{E}[\varphi(X) | Y = y] &= \int \varphi(x) \mathbb{P}(X \in dx | Y = y) = \int \varphi(x) f_{X|Y}(x|y) dx && \text{(conditional expectation of } \varphi(X) \text{ given } Y = y) \\
- \mathbb{E}[\varphi(X)] &= \mathbb{E}[\mathbb{E}[\varphi(X) | Y = y]] = \int \mathbb{E}[\varphi(X) | Y = y] \mathbb{P}(Y \in dy) && \text{(expectation of } \varphi(X))
\end{aligned}$$

Exercises

Example 3.1.7 (Gamma random variables)

Let $X \sim \text{Gamma}(k, \mu)$. We define Gamma random variables in the by specifying their CDF $\mathbb{P}(X \leq x) = 1 - \sum_{n=0}^{k-1} e^{-\mu x} (\mu x)^n / n!$. Show that the density of X is

$$\frac{d}{dx} \mathbb{P}(X \leq x) = \frac{\mu e^{-\mu x} (\mu x)^{k-1}}{(k-1)!}$$

and that $\mathbb{E}[X] = k/\mu$.

Exercise 1 (Not All Random Variables have names)

Suppose we have a random variable X for which $P(X > x) = 1 - x^4$, $x \in [0, 1]$. Find the expectation, variance, and density function of X .

Exercise 2 (Betting on fedoras)

Having acquired some 309 wisdom, Jessica, together with her ORF friends (all over 21), headed to Atlantic City over spring break to have some fun and make some money. The city has been way past its glory days for quite some time now, and as far as the casino business goes could certainly use a breath of fresh air. No wonder our heroes get instantly bored and go to the closest coffee shop to regroup and re-think their spring break plans. As they are sipping their coffee and staring at the forming line, it becomes clear to Jessica that wearing fedoras is a thing in this city. After a multitude of observations she realizes that a fedora-wearing person is equally likely to arrive at any point in time and on average there is a fixed number μ of independent fedora-wearers arriving every hour.

- a) Jessica offers the group to play the following game: throw a fair die, and record the time when the first fedora-wearer walks in. For every dot on the die Jessica will receive 10 dollars, and for every minute on the clock, she will pay 1 dollar. Jessica is an honest person, so the game she has offered is perfectly fair. How much time does it take on average for a fedora-wearer to arrive? What is the variance of Jessica's return?

- b) Let's change the game up a bit: now after recording the outcome of the dice and the arrival time Jessica flips an independent coin, and if it comes up heads the arrival time gets erased, and if tails, the die-outcome gets erased. This new game is assumed to be perfectly fair. In this case, how much time does it take on average for a fedora-wearer to arrive? What is the variance of Jessica's return? Give the cdf of the amount of money that changes hands.

Exercise 3 (Forest Fire)

The number of forest fires in California in a given year is a random variable. This number depends in part on the length of the drought season. Assume that the length of the drought season X is exponential with parameter μ . Given $X = x$ assume that the number of fires is conditionally Poisson with mean x . Find the probability that there are at least three fires this season.

Exercise 4 (Fish Delivery)

Jerry has decided to open a new sushi restaurant "Prince-Udon" in Princeton, and since there are no affordable fish farms in the local area he decides to get his supplies from the fish market in Philly. Fortunately for him, a new app, FoodEx, provides ingredient delivery services for small restaurant owners. Before getting to Jerry the fish will travel to a distribution center A in between Princeton and Philly. Being a meticulous entrepreneur, Jerry wants to know how much time it takes for the fish to get from Philly to the distribution center and from the distribution center to Princeton. FoodEx obliges and tells Jerry that the joint density of Philly to A travel time X and the total travel time Y is given by

$$P\{X \in dx, Y \in dy\} = f_{X,Y}(x, y) dx dy = \lambda^2 e^{-\lambda y} dx dy$$

for $0 \leq x \leq y < \infty$. What are the marginal densities of the total and Philly to A travel times? Find the conditional distribution of total travel time given that the travel time from Philly to A is x . Find the distribution of the travel time from A to Princeton.